From Norm-Five Quaternions to the Ramanujan Circle: Complete Nonbacktracking Dynamics of the LPS $X^{5,q}$ Family

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Abstract

For every prime $q > 5$ with $q \equiv 1 \pmod 4$, six Hamilton quaternions of norm five define a six-regular Lubotzky–Phillips–Sarnak Cayley graph. We close its arithmetic and dynamical specialization in one normalization. The Legendre symbol $(5/q)$ selects a connected $\mathrm{PSL}_2(\mathbb{F}_q)$ or $\mathrm{PGL}_2(\mathbb{F}_q)$ chamber and exactly controls bipartiteness. The oriented-edge Hashimoto operator has a complete Bass characteristic factorization, all-iterate trace formula, Möbius primitive-cycle inversion, and finite-graph Ihara product. The LPS Ramanujan bound then places every nontrivial quadratic Hashimoto root on $|\mu| = \sqrt{5}$; the two chambers each have conditional prime density 1/2. Five full finite groups provide exact regression evidence, not a proof by sampling. This is a source-arithmetic Route-A exploration, with no target Euler product, zero correspondence, or Hilbert–Pólya claim.

Keywords: LPS graphs; arithmetic dynamics; Hashimoto operator; primitive cycles; Ramanujan spectrum; prime number theorem; computational certification

中文摘要

我们从范数五的六个整数四元数出发，把它们约化为有限域上的投影矩阵，并冻结右乘的Cayley规范。若素数 $q > 5$ 且 $q \equiv 1 \pmod 4$，则平方类 $(5/q)$ 把同一构造分入 $\mathrm{PSL}_2(\mathbb{F}_q)$ 与 $\mathrm{PGL}_2(\mathbb{F}_q)$ 两个算术室。前者非二分，后者由行列式平方类给出等大的二分部；群阶、六度性以及更换 $\sqrt{-1}$ 所引起的共轭等价均被明确确定。 在此图上提升到有向边后，禁止立即逆行的转移给出一个内生动力学。我们专门化Bass–Hashimoto行列式，固定所有次幂迹的闭步约定，并用Möbius反演恢复每个长度的本原有向闭轨；相应的有限图Ihara乘积因而逐项确定。 再调用LPS谱界，非平凡二次根恰落在半径 $\sqrt{5}$ 的圆上；算术级数素数定理还给出两室各为二分之一的条件自然密度。五个完整有限群只用作精确实现凭据。错余类、合数和室标签置换已完成A0对照，但不能替代本原轨道层的周期、相位、权重及等密度长度对照；模数也没有转移为轨道标签。严格评估故记A1_WEAK，仅报告路线A探索，不声称目标Euler因子、零点对应或Hilbert–Pólya算子。

关键词： 算术动力学；LPS图；非回溯算子；本原闭轨；Ramanujan谱；算术级数素数定理；计算机核验

1 Arithmetic quaternion chambers

Let

$$\mathcal{S}_5 = \{(1, \pm 2, 0, 0), (1, 0, \pm 2, 0), (1, 0, 0, \pm 2)\}.$$

Choose $\iota \in \mathbb{F}_q$ with $\iota^2 = -1$ and put

$$M_\iota(a) = \begin{pmatrix} a_0 + \iota a_1 & a_2 + \iota a_3 \\ -a_2 + \iota a_3 & a_0 - \iota a_1 \end{pmatrix}. \tag{1}$$

We use its projective class. Write $\chi_q = (5/q)$ and

$$G_q = \begin{cases} \mathrm{PSL}_2(\mathbb{F}_q), & \chi_q = 1, \\ \mathrm{PGL}_2(\mathbb{F}_q), & \chi_q = -1. \end{cases}$$

Lemma 1 (Norm, inverses, and gauge)**.** *The six classes $[M_\iota(a)]$ are distinct and inverse closed. Replacing ι by $-\iota$ conjugates the generating set and gives an isomorphic labeled Cayley dynamics.*

Proof. Expansion gives

$$\det M_\iota(a) = a_0^2 + a_1^2 + a_2^2 + a_3^2 = 5.$$

Quaternion conjugation changes the sign of the unique nonzero tail coordinate and satisfies $M(a)M(\bar{a}) = 5I$. The paired projective classes are inverses. For $q > 5$, equality of two classes, or equality with the identity class, would force one of $2, 4, 5$ to vanish modulo q. Thus the six classes are distinct and nonidentity. For $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, direct multiplication gives $M_{-\iota}(a) = JM_\iota(a)J^{-1}$. □

Theorem 1 (LPS chamber closure)**.** *The Cayley graph $X^{5,q} = \mathrm{Cay}(G_q, \mathcal{S}_5)$ is connected, simple, six-regular, and Ramanujan. More precisely,*

$$|G_q| = \begin{cases} q(q^2-1)/2, & \chi_q = 1, \\ q(q^2-1), & \chi_q = -1. \end{cases}$$

The first chamber is nonbipartite. The second is bipartite, with the square and nonsquare projective determinant classes as its two equal parts.

Proof. Connectedness, the group identification, and the Ramanujan bound are the LPS theorem specialized to quaternion prime five [1]. Lemma 1 gives simplicity and degree six, while the displayed orders are the standard projective-group orders. Projective determinant square class is well defined because scalar rescaling changes determinant by a square. If $\chi_q = -1$, every edge flips this character; connectedness and one generator give a bijection between the two parts. If $\chi_q = 1$, a connected bipartite Cayley graph would define a nontrivial word-parity map $\mathrm{PSL}_2(\mathbb{F}_q) \to \{\pm 1\}$. Its kernel would have index two, contradicting the simplicity of $\mathrm{PSL}_2(\mathbb{F}_q)$ for $q \geq 13$. □

2 Exact primitive nonbacktracking dynamics

Let $n_q = |G_q|$ and $m_q = 3n_q$. On the $2m_q = 6n_q$ oriented edges, define $H_q(e, e') = 1$ when the terminus of e is the origin of e' and $e' \neq \bar{e}$. Let A_q be adjacency on vertices.

Theorem 2 (Complete determinant and orbit ledger)**.** *One has the polynomial identities*

$$\det(tI - H_q) = (t^2 - 1)^{2n_q} \det(t^2 I - tA_q + 5I), \tag{2}$$

$$\det(I - uH_q) = (1 - u^2)^{2n_q} \det(I - uA_q + 5u^2 I). \tag{3}$$

If $N_q(r) = \mathrm{Tr}(H_q^r)$ and $\Pi_q(r)$ counts prime oriented cycles of length r, modulo cyclic shift but not reversal, then

$$\Pi_q(r) = \frac{1}{r} \sum_{d|r} \mu(d) N_q(r/d). \tag{4}$$

Moreover, in $\mathbb{Q}[[u]]$ and as a rational function,

$$\prod_{[C]} (1 - u^{\ell(C)})^{-1} = \exp\left(\sum_{r \geq 1} \frac{N_q(r)}{r} u^r \right) = \det(I - uH_q)^{-1}. \tag{5}$$

Proof. The Bass–Hashimoto formula for a graph with degree matrix D is

$$\det(I - uH) = (1 - u^2)^{m-n} \det(I - uA + u^2(D - I)).$$

Here $D = 6I$ and $m - n = 2n$, which proves (3); substitution $u = t^{-1}$ proves (2) [2, 3, 4]. By construction, a diagonal entry of H_q^r selects a closed oriented-edge walk with no inverse pair, including across the terminal seam. Thus its trace counts cyclically nonbacktracking walks. Every such walk is a unique repeat of a prime oriented cycle, so $N_q(r) = \sum_{d|r} d\Pi_q(d)$. Möbius inversion proves (4); grouping all repeats, or taking the formal logarithm of the finite determinant, proves (5). □

3 Ramanujan circle and route boundary

Theorem 3 (Complete Hashimoto spectral placement)**.** *For every $\lambda \in \mathrm{Spec}(A_q)$, the two roots assigned by* (2) *solve $\mu^2 - \lambda\mu + 5 = 0$. The factor $(t^2-1)^{2n_q}$ adds $+1$ and -1, each with multiplicity $2n_q$. The trivial adjacency root* 6 *gives* 5, 1*. The root -6 gives $-5, -1$ and occurs exactly in the bipartite* PGL_2 *chamber. Every other quadratic root obeys $|\mu| = \sqrt{5}$.*

Proof. Equation (2) is a complete factorization of degree $6n_q$. The LPS Ramanujan theorem gives $|\lambda| \le 2\sqrt{5}$ after removing 6 and the possible bipartite root -6. In the strict case the quadratic roots are conjugate with product five; at equality they coincide at $\pm\sqrt{5}$. Hence both have modulus $\sqrt{5}$. The two trivial factorizations are $(\mu-1)(\mu-5)$ and $(\mu+1)(\mu+5)$. A connected regular graph has eigenvalue minus its degree exactly when it is bipartite, and Theorem 1 identifies that chamber. ☐

Proposition 1 (Prime chamber density)**.** *Among primes $q > 5$ with $q \equiv 1 \pmod 4$, the* PSL_2 *and* PGL_2 *chambers each have conditional natural density* 1/2*. They are respectively $q \equiv 1, 9 \pmod{20}$ and $q \equiv 13, 17 \pmod{20}$.*

Proof. Quadratic reciprocity gives $(5/q) = (q/5)$. The four displayed reduced classes modulo 20 each have natural density 1/8 among all primes by the prime number theorem for arithmetic progressions [5]. Conditioning on their union gives weight 1/4 per class and 1/2 per pair. ☐

4 Exact evidence, limitations, and declarations

The executable certificate constructs the complete groups for $q = 13, 17, 29, 37, 41$, totaling 104,316 vertices and 625,896 oriented edges. It verifies 60 trace rows through time 12. Independent code uses a different projective normalization and the opposite multiplication side. The certified girths are respectively $8, 8, 9, 10, 9$. Among the 1,124 eligible primes at most 20,000, the finite chamber counts are 555 and 569; these are regression data, not a density estimate.

The arithmetic construction clears A0. The exact primitive ledger is source-local: it does not turn the modulus q or its primality into a label of an individual orbit, establish $p \leftrightarrow \gamma_p$ or prime-power repetition, or produce intrinsic $\log p$ or von Mangoldt weights. The A0 lane executes wrong-residue-prime, matched-composite, and cyclically shuffled chamber-label controls. These do not replace the A1 controls: mandatory shuffled-period, random-weight, random-phase, same-density-length, neighboring-parameter, and simpler-parent orbit-correspondence controls are absent. Strict evaluator v0.2 therefore records A1 as weak, not as a pass. HCS-C329 already owns the generic Bass/Ihara/Hashimoto mechanism; C375 claims only this LPS-specific joint atlas. The separate finite graphs do not form one autonomous global flow, and no target determinant or divisor is identified. Consequently A2 and A3 fail, while the canonical adjacency and nonnormal Hashimoto operators supply only an A4 formal hint. The tuple is

$$(A0_{\mathrm{STRUCTURAL\_ARITHMETIC\_RELATION}}, A1_{\mathrm{WEAK}}, A2_{\mathrm{FAIL}}, A3_{\mathrm{FAIL}}, A4_{\mathrm{FORMAL\_HINT}}).$$

The overall status is `ROUTE_A_EXPLORATORY` under `NO_BAD_EULER_OR_ROOT_NUMBER`. There is no target arithmetic local datum, target Euler product, bad-prime datum, root number, automorphy, target functional equation, target zero match, target counting law, or Hilbert–Pólya operator. Route B is not invoked.

Data availability. All exact evidence and source code are included in the C375 package.

Ethics declaration. This mathematical study uses no human participants, animals, or sensitive personal data.

Author contributions (CRediT). Conceptualization, formal analysis, software, validation, writing, and revision are recorded as one auditable package; final responsibility remains with the submitting author.

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AI-use disclosure. Generative AI assisted drafting and code generation. The package exposes the proof dependencies, exact computations, independent checker, and hostile tests so that the authors can verify every released claim.

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